

JIAMING ZENG

jiaming@alumni.stanford.edu | Union City, CA | Google Scholar | Website | 901-338-4651

EDUCATION

Stanford University

- M.S & Ph.D. in Management Science and Engineering

Stanford, CA

Sept. 2016 – June 2021

Massachusetts Institute of Technology

- B.S. in Mathematics with Computer Science
- Minor in Applied International Studies/Concentration in German

Cambridge, MA

Feb 2015

WORK EXPERIENCE

Google DeepMind, Personalization

Mountain View, CA

Senior Research Engineer / Tech Lead

Dec 2025 - Present

- Architected, prototyped, and launched live experiments for "Memory Chains" in long-term memory, establishing automated and human evaluation baselines to sustain multi-turn conversational continuity over extended horizons and define the research and production roadmap.
- Optimized inference efficiency through log-driven triggering analysis and TPU-optimized model migrations, cutting redundant computations to save 1,300+ GXUs and earning an organizational TPU Harvester Award.
- Built e2e LLM evaluation infrastructure—including automated Side-by-Side (SxS) benchmarking harnesses, a Chain-of-Thought reasoning trace analyzer, autoraters, and human evaluation frameworks—to diagnose reasoning trajectories and gate model releases.
- Standardized offline configuration management across distributed microservices by replacing static flows with a centralized, flag-driven system, cutting launches from weeks to 1–2 days and unblocking multiple global rollouts.
- Directed team-wide model quality operations and launch gating, co-authored research proposals on long-context LLM memory architectures for next-generation frontier models, and mentored engineers to ensure experimental rigor across high-velocity release cycles

Google, GCP Analytics AI

Sunnyvale, CA

Senior Machine Learning Engineer / Tech Lead

June-2024 - Nov 2025

- Drove e2e delivery and technical coordination of BigQuery Conversational Analytics from prototype to private preview in Nov 2025 and defining the North Star vision to guide future engineering scaling and development.
- Increased BigQuery user-acceptance rate by 13.3% and executability by 11.3% through self-consistency.
- Pioneered human-in-the-loop (HITL) work for disambiguation and explainability on agentic workflows.
- Led Looker Query generation and delivering 27.8% accuracy and 7% executability improvements; owning service reliability, long-term strategy, and the migration to Java ADK.

AKASA

South San Francisco, CA

Staff Machine Learning Researcher / AI Tech Lead

Sept. 2022 - June 2024

- Lead R&D for human-in-the-loop AI systems for healthcare operations from inception to business impact.
- Researched techniques for extrapolating LLMs to long contexts on LongT5, MPT-7B, Llama2-7/13B, etc.
- Built LLM pipeline from infrastructure, data processing, continued pretraining, fine-tuning, evaluation, to launch.

X, The Moonshot Factory [Google X]

Mountain View, CA

AI Residency

June 2019 – Sept. 2019

- Developed technology to protect the ocean and improve sustainable fishing
- Delivered ML models to gain insights into fishing practices from limited data

NVIDIA, AI Infrastructure

Santa Clara, CA

AI Research Intern

June 2018 – Sept. 2018

- Optimized Bayesian neural networks to capture model uncertainty in active learning (NeurIPS 2018) ([paper](#))
- Contributed Bayesian neural network training code to the official TensorFlow Probability [repository](#)

Oracle, Storage SW Quality Engineering

Burlington, MA

Software Engineer

Mar. 2015 – Sept. 2016

- Optimized machine learning models for the quality assurance process
- Ensured the quality and stability of the ZFS Storage Appliance device

NOAA Southwest Fisheries Science Center

San Diego, CA

Ernest E. Hollings Scholar

May 2014 – Aug. 2014

- Contributed to development of machine vision app in C# for fish detection and tracking in underwater videos
- Paper published in the IEEE Applications and Computer Vision Workshop 2015 ([doi:10.1109/WACVW.2015.11](#))

BMW, R&D ConnectedDrive Group

Munich, Germany

Software Engineer

May 2013 – Aug. 2013

- Programmed a particle filtering system for the driver intent inference system to predict the driver's gaze direction
- Designed a calibration GUI for SmartTrack, a device used in data gathering for driver intent inference

RESEARCH EXPERIENCE

IBM Research, Computational Health Group

Cambridge, MA

Postdoctoral Researcher

Sept. 2021 – Sept. 2022

- Researched AI fairness and bias in healthcare treatments through analysis of clinical notes in MIMICIII ([paper](#))
- Collaborated across industry and academia to research fairness through synthetic healthcare data ([paper](#))

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- Researched methods for search and retrieval on clinical documents with word embeddings and SBERT

Stanford Graduate School of Business, Athey Lab

Stanford, CA

Advisor: Prof. Susan Athey

June 2017 – June 2021

- Built interpretable AI tools to reduce selection bias on causal inference for cancer treatment studies
- Researched RL, Gaussian Process regression, and Bayesian optimization for parameter tuning on causal forests
- Simulated causal inference datasets and experiments to test various parameter tuning methods

Stanford, Management Science and Engineering, Decision Analysis Group

Stanford, CA

Advisor: Prof. Ross Shachter

Sept. 2016 – June 2021

- Modeled a radiologist's decision threshold for mammography screening with Bayesian networks
- Developed decision support tool for oncology decision making and treatment support
- Paper published to *Medical Decision Making* (doi: 10.1177/0272989X19832914)

Stanford, School of Medicine, Laboratory of Quantitative Imaging and Artificial Intelligence

Stanford, CA

Advisor: Prof. Daniel Rubin

Sept. 2018 – June 2021

- Trained NLP embedding models to identify and extract treatment information from EMR data ([student abstract](#))
- Project funded by the Stanford Human-Centered Artificial Intelligence Institute Seed Grant ([proposal](#))

MIT Sloan School of Management, Prediction Analysis Lab

Cambridge, MA

Advisor: Prof. Cynthia Rudin

Feb. 2014 – Feb. 2015

- Developed interpretable models using machine learning algorithms to predict the probability of prisoner recidivism
- Mined the 1994 Prisoner Recidivism data for essential features that can be used in model development
- Paper published in the *Journal of Royal Statistical Society, Series A* (doi: 10.1111/rssa.12227)

MIT, Department of Mechanical Engineering

Cambridge, MA

Advisor: Prof. Alexie Kolpak

Sept. 2012 – May 2013

- Developed a theoretical “green” engine model that produces its own fuel by converting CO₂ into methanol
- Analyzed theoretical processes for heat transfer during carbon dioxide to methanol reaction

Tsinghua University, Department of Computer Science

Beijing, China

Advisor: Prof. Wang Xiaoge

June 2012 – Aug. 2012

- Optimized the performance of the Conjugate Gradient Solver in Los Alamos Lab's Parallel Ocean Program (POP)
- Operated experiments on supercomputers such as Tianhe1A and BlueLight through remote control

AWARDS, GRANTS, & SCHOLARSHIPS

- Stanford Institute of Human-Centered Artificial Intelligence Seed Grant Recipient (2019)
- 1st Place, USPROC Undergraduate Statistics Research Project Competition (2015)
- NOAA Ernest E. Hollings Scholarship (2013-2015)

PUBLICATIONS

- Bhanot, K., Baldini, I., Wei, D., **Zeng, J.**, & Bennett, K. (2023, August). Stress-Testing Bias Mitigation Algorithms to Understand Fairness Vulnerabilities. In *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 764-774).
- **Zeng, J.**, Gensheimer, M. F., Rubin, D. L., Athey, S., & Shachter, R. D. (2022). Uncovering interpretable potential confounders in electronic medical records. *Nature Communications*, 13(1), 1014.
- Adam, H., Yang, M. Y., Cato, K., Baldini, I., Senteio, C., Celi, L. A., **Zeng J.**, Singh M & Ghassemi, M. (2022, July). Write it like you see it: Detectable differences in clinical notes by race lead to differential model recommendations. In *Proceedings of the 2022 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 7-21).
- **Zeng, J.**, Banerjee, I., Henry, A. S., Wood, D. J., Shachter, R. D., Gensheimer, M. F., & Rubin, D. L. (2021). Natural language processing to identify cancer treatments with electronic medical records. *JCO Clinical Cancer Informatics*, 5, 379-393.
- **Zeng, J.** (2020). Developing a Machine Learning Tool for Dynamic Cancer Treatment Strategies. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(10), 13742-13743.
- **Zeng, J.**, Gimenez, F., Burnside, E. S., Rubin, D. L., & Shachter, R. (2019). A probabilistic model to support radiologists' classification decisions in mammography practice. *Medical Decision Making*, 39(3), 208-216.
- **Zeng, J.**, Lesnikowski, A., & Alvarez, J. M. (2018). The Relevance of Bayesian Layer Positioning to Model Uncertainty in Deep Bayesian Active Learning. In *3rd Annual Bayesian Deep Learning Workshop, NeurIPS 2018*.
- **Zeng, J.**, Ustun, B., & Rudin, C. (2017). Interpretable classification models for recidivism prediction. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 180(3), 689-722.
- Cutter, G., Stierhoff, K., & **Zeng, J.** (2015). Automated detection of rockfish in unconstrained underwater videos using Haar cascades and a new image dataset: labeled fishes in the wild. In *Applications and Computer Vision Workshops (WACVW), 2015 IEEE Winter* (pp. 57-62). IEEE.
- [Non-academic] **Zeng, J.** (2019). My day as a double major. *Firehose, MIT Technology Review*. March 2019.

LANGUAGES

English (Native), Chinese (Native), German (Advanced), Python, PyTorch, R, Java, Golang, Kotlin, MATLAB, C#, C++, LaTeX, bash

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LEADERSHIP & TEACHING

Decision Analysis Working Group Seminar

Seminar Organizer

Stanford, CA

Jan. 2020 – June 2021

Women in Mathematics, Science, and Computational Engineering (WiMSCE)

Events Chair

Stanford, CA

Sept. 2019 – Jun 2020

MS&E 152/252: Introduction to Decision Analysis

Course Assistant for Prof. Ross Shachter

Stanford, CA

Spring 2017, 2019; Winter 2021

MS&E 120/220: Probabilistic Analysis

Course Assistant for Prof. Ross Shachter

Stanford, CA

Fall 2018